Return Trip

Team 03 · Eranova hackathon · Ryan (backend & AI agent) · Asher (email pipeline) · Ali (frontend) · Nov 13, 2026

One line: an AI agent on top of the broker's TMS that finds where trucks drive home empty and fills those trips, with a human approving every write. *Eranova makes every load run smoothly; Return Trip makes sure the truck never drives home empty.*

85,556	49%	99%	$194k	160
empty miles in Meridian's history	trucks loaded today	of empty miles on 15 one-way trips	spent driving empty (at $2.27/mi)	real loads synced + AI-classified

The problem

Meridian books every load one at a time and its freight flows one way (Dallas receives 37 loads, ships 0). After delivering, almost every truck drives home empty and someone pays for that trip. Replaying all 130 truckloads showed that rearranging Meridian's own loads saves only 0.7%: the fix is winning new freight for the trip home.

Our solution

- **Finds the empty trips:** network analysis of real Meridian loads; the 15 one-way trips become sales targets with real leads.
- **Fills them:** finds compatible return freight (equipment, FSMA food-after-hazmat rule, hazmat, size, carrier compliance), prices it, and queues a real Meridian quote.
- **Chains trucks into loops** where real loads allow, and shows one loop to the broker, the carrier and the manufacturer.
- **Learns from history:** dock waits, on-time rates, fastest carriers, lead times. **A human approves every write to Meridian.**

What each of us built

Who	Built	Status
Ryan Backend & AI brain	Flask API; Meridian client; sync of 160 shipments, 18 carriers, 30 facilities, 14 customers; GPT-4o freight classification; deterministic compatibility rules; network + backhaul finder with scoring; backhaul pricing; approval queue that creates real quotes; GPT-4o tool-calling agent; demo reset script; offline test suite.	Merged and live. 69/69 tests pass. Agent found and queued a Dallas → Chicago backhaul on its own.
Asher Email pipeline	Email intake: parse → AI classify (quote request, washout certificate, last-load info, carrier offer) → AI extraction → match to Meridian and return trips → draft quote + draft reply queued for approval; washouts table that unblocks the FSMA rule.	Designed and in progress; not merged yet. The UI already shows each email's actions with approve/reject.
Ali Frontend	Map-first app with a 3D map (real road routes, direction arcs, status colours). **Broker:** loops ready to book, empty trips to fill, my loops with money in / out / margin / miles saved. **Carrier:** fleet + driver phone. **Manufacturer:** date-shift offers. Weekly reports with charts, learned rules, inbox actions, agent chat. Planner + learning scripts on real data.	Done, PR #3 open. Runs on Ryan's live API; falls back to real-data files for routes not built yet.

How we use AI

- **AI agent (live):** GPT-4o calls tools step by step (network overview, find backhauls, check rules, price) and queues proposals for human approval.
- **AI classification (live):** every shipment sorted into food, hazmat, steel, etc.; the food-safety rule depends on it.
- **AI email understanding (in progress):** reads quote requests and carrier emails into structured freight and draft replies.
- **Rules and statistics decide, AI advises:** compatibility rules are code, not AI; loop planning and learning are algorithms on real history.

Data and next steps

All numbers come from real Meridian data (loads, trucks, prices, on-time records, rate-engine market rates, OpenStreetMap routes). Only assumption: $2.27 per empty mile (ATRI industry average). **Next:** merge Asher's email pipeline; port the loop planner, return lanes and reports into backend routes; make booking a loop cover its loads in Meridian.